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Integrative genomic analysis in African American children with asthma finds three novel loci associated with lung function

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Abstract

Bronchodilator (BD) drugs are commonly prescribed for treatment and management of obstructive lung function present with diseases such as asthma. Administration of BD medication can partially or fully restore lung function as measured by pulmonary function tests. The genetics of baseline lung function measures taken before BD medication have been extensively studied, and the genetics of the BD response itself have received some attention. However, few studies have focused on the genetics of post-BD lung function. To address this gap, we analyzed lung function

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AUTHOR CONTRIBUTIONS

P. C. G., K. L. K., and M. J. W. designed the study. S. S. O., S. S., C. E., and E. G. B. recruited study subjects and generated the data. A. C. Y. M., J. R. E., D. U., and S. H. cleaned and organized the data and provided analytic support. P. C. G., K. L. K., E. Y. L., O. R., M. G. C., and M. J. W. performed the analysis. A. K. L. and L. S. B. provided clinical pharmacological expertise for interpretation of results. E. G. B. and B. E. H. funded the study. E. G. B. supervised all recruitment. All authors contributed to manuscript writing and editing.

CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in the NCBI dbGaP repository under accession number phs000921.v1.p1.

SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

phenotypes in 1103 subjects from the Study of African Americans, Asthma, Genes, and Environment, a pediatric asthma case-control cohort, using an integrative genomic analysis approach that combined genotype, locus-specific genetic ancestry, and functional annotation information. We integrated genome-wide association study (GWAS) results with an admixture mapping scan of three pulmonary function tests (forced expiratory volume in 1 s [FEV₁], forced vital capacity [FVC], and FEV₁/FVC) taken before and after albuterol BD administration on the same subjects, yielding six traits. We identified 18 GWAS loci, and five additional loci from admixture mapping, spanning several known and novel lung function candidate genes. Most loci identified via admixture mapping exhibited wide variation in minor allele frequency across genotyped global populations. Functional fine-mapping revealed an enrichment of epigenetic annotations from peripheral blood mononuclear cells, fetal lung tissue, and lung fibroblasts. Our results point to three novel potential genetic drivers of pre- and post-BD lung function: *ADAMTS1*, *RAD54B*, and *EGLN3*.

Keywords

admixture; African American; asthma; GWAS; integrative genomic analysis; lung function

1 | INTRODUCTION

Asthma is a disease characterized by episodic obstruction of airways that affects nearly 339 million people worldwide (The Global Asthma Network, 2018) and is the most common chronic disease among children. Asthma constitutes a massive global economic burden, representing \$81.9 billion in medical costs in the United States alone (Nurmagambetov et al., 2018). As a complex disease, asthma results from both environmental and genetic factors, with genetic heritability estimates ranging from 0.35 to 0.90 (Ober & Yao, 2011). The advent of genome-wide association studies (GWAS; Risch & Merikangas, 1996), combined with progressively larger sample sizes in recent years, has enabled researchers to query the genetic basis of asthma at unprecedented scale, with numerous loci identified in autoimmune and inflammatory pathways (Demenais et al., 2018). However, these loci account for a small portion of asthma liability (Demenais et al., 2018).

Pulmonary function tests are recommended to guide the diagnosis of asthma and monitor patient status (Asthma and Allergy Foundation of America, 2019). During these tests, patients breathe through a spirometer that captures key measures of lung function, including the forced expiratory volume in 1 s (FEV₁), which measures initial forced exhalatory capacity; the forced vital capacity (FVC), which measures the maximum total volume of air that a patient can forcibly exhale; and their ratio (FEV₁/FVC). Lung function measures can be population-normalized according to expected lung function values that account for age, sex, height, and ethnicity of the patient (Hankinson et al., 1999). Spirometric measurements can be taken both before bronchodilator (BD) treatment (pre-BD) and after (post-BD) to further understand lung function status. Historically, baseline lung function is measured with pre-BD measures, but among people with asthma, post-BD lung function may best reflect lung health (Brehm et al., 2015).

While the genetic contribution to asthma and lung function has been extensively studied via GWAS, most analyses have relied on subjects of European descent (Demenais et al., 2018; Johansson et al., 2019; Pickrell et al., 2016; Z. Zhu et al., 2018). This overrepresentation of ethnically white subjects in biomedical research has impaired the generalizability of genetic studies of complex disease (Burchard, 2014; Bustamante et al., 2011; Popejoy & Fullerton, 2016). Ethnic differences in lung function, particularly between non-Hispanic African Americans and European Americans, have been reported for over 40 years (Binder et al., 1976; Glindmeyer et al., 1995; Hsi et al., 1983; Rossiter & Weill, 1974; Schwartz et al., 1988). Ethnic disparities in lung function were attributed to population differences in sitting height, as increased height leads to increased lung capacity. However, adjustment for sitting height only explains 42%–50% of ethnic differences in lung function between African Americans and European Americans (Harik-Khan et al., 2004), suggesting that a simplistic reduction to ethnic differences in height cannot account for the observed disparity in lung function. Unequal socioeconomic conditions were also thought to contribute to ethnic differences in lung function (Braun, 2015; Quanjer, 2013, 2015), but socioeconomic factors only account for 7%–10% of unexplained variance (Harik-Khan et al., 2004). Self-identified race or ethnicity are commonly used in the clinic to interpret lung function measures, but these are not ideal variables for understanding *genetic* differences in lung function between populations. Kumar et al. (2010) observed that the proportion of global African genetic ancestry is inversely correlated with lung function. Spear et al. (2019) later observed population differences among African Americans, Mexican Americans, and Puerto Ricans in BD drug response to albuterol, the short-acting β_2 -adrenergic receptor agonist that is the most commonly prescribed drug for the treatment of acute asthma symptoms. Specifically, Spear et al. performed admixture mapping, a technique designed to identify regions of the genome where locus-specific ancestry drives variation in a disease trait (Shriner, 2013) that has been helpful in studies of complex diseases, including asthma and breast cancer (Féjerman et al., 2012; Pino-Yanes et al., 2015). However, admixture mapping studies comparing baseline and post-BD lung function have not yet been performed in African Americans. In this study, we address this gap in knowledge by evaluating the effect of locus-specific ancestry on both pre- and post-BD lung function measures in a pediatric case–control cohort of non-Hispanic African Americans children and adolescents.

2 | METHODS

2.1 | Ethics statement and data availability

Data from the Study of African Americans, Asthma, Genes, and Environments (SAGE) cohort were used for this study. The data that support the findings of this study are available in the NCBI dbGaP repository under ascension number phs000921.v1.p1. Data from SAGE were approved for human subjects research under expedited review by IRB 10–02877 at the University of California, San Francisco with reference #244919. All subjects gave written consent for genotyping, phenotyping, and data usage for general research use.

2.2 | Study population

SAGE is a case–control cross-sectional cohort study of genetics and gene–environment interactions in non-Hispanic African American children and adolescents in the United

States. SAGE includes detailed clinical, social, and environmental data on both asthma and asthma-related conditions. Full details of the SAGE study protocols are described in detail elsewhere (Borrell et al., 2013; Mak et al., 2018; Nishimura et al., 2013; Thakur et al., 2013). Briefly, SAGE was initiated in 2006 and recruited participants with and without asthma through a combination of clinic- and community-based recruitment centers in the San Francisco Bay Area. All participants in SAGE self-identified as African American and self-reported that all four grandparents were non-Hispanic African American. To reduce population heterogeneity resulting from very recent admixture, we only analyzed subjects who affirmatively self-identified as having been born in the United States and whose parents were also born in the United States.

Pulmonary function tests were taken before administration of albuterol BD medication for all individuals, both those with and without asthma. Post-BD spirometry measures were performed only for individuals with asthma. Analyses of pre-BD lung function measures included all 1103 asthma cases and controls with complete covariate information. Post-BD analyses were performed on the 831 asthma cases with post-BD measurements.

2.3 | Genotyping and quality control

DNA was isolated from whole blood collected from SAGE participants at the time of study enrollment as described previously (Borrell et al., 2013). DNA was extracted using the Wizard® Genomic DNA Purification kits (Promega). Samples were genotyped with the Affymetrix Axiom LAT1 array (World Array 4, Affymetrix).

Genotype quality control was performed in PLINK v1.9 (Chang et al., 2015). Of the 772,703 genotyped variants, 111,901 single nucleotide polymorphisms (SNPs) were excluded from analysis due to genotype missingness more than 5% ($n = 28,211$), minor allele frequency (MAF) less than 1% ($n = 80,420$) or deviation from Hardy-Weinberg expectations (HWE) at $p < 0.001$ ($n = 3270$). The final set included 660,802 genotyped markers (Table S1).

Genotyped SNPs were submitted to the Michigan Imputation Server (Das et al., 2016), phased using EAGLE v2.3 (Loh et al., 2016), and imputed from the 1000 Genomes Project reference panel (The 1000 Genomes Project Consortium, 2015) using Minimac3 (Das et al., 2016). Imputed SNPs with imputation $R^2 < .3$, with deviation from HWE $p < 10^{-4}$, or with MAF $< 1\%$ were discarded. Of the 47,101,126 imputed SNPs, a total of 31,146,322 were culled due to either low MAF ($n = 31,095,418$) or deviation from HWE ($n = 50,904$). All variants in the imputed set showed a genotype missingness of no more than 5%. The final number of SNPs used in association analyses was 15,954,804 (Table S1).

2.4 | Outcome phenotypes

Pulmonary function testing was performed at the time of recruitment according to the American Thoracic Society/European Respiratory Society standards (Miller et al., 2005; Pellegrino et al., 2005; Wanger et al., 2005) with a KoKo PFT Spirometer (nSpire Health Inc.). Spirometry was performed both before and 15 min after administration of four puffs of albuterol (90 µg per puff) through a 5-cm plastic mouthpiece from a standard metered-dose inhaler. Patients were assessed for the following spirometric measures before and after BD drug usage (pre-BD and post-BD, respectively): (a) FEV₁, (b) FVC, and (c) FEV₁/FVC. A

total of six phenotypes were assessed for genotype association: pre-BD FEV₁ (pre-FEV₁), pre-BD FVC (pre-FVC), pre-BD FEV₁/FVC (pre-FEV₁/FVC), post-BD FEV₁ (post-FEV₁), post-BD FVC (post-FVC), and post-BD FEV₁/FVC (post-FEV₁/FVC). All phenotype values were normalized based on the expected lung function values calculated from the Hankinson equations (Hankinson et al., 1999), which account for age, sex, height, and self-reported ethnicity. Phenotype distributions were checked for normality and to detect outliers. Outliers were determined using the method of Tukey fences (John Tukey, 1977). For each phenotype, we computed the first quartile value (Q1), the third quartile value (Q3), and the interquartile range (IQR). We declared as outliers all values outside of the range

$$[Q1 - 3(IQR), Q3 + 3(IQR)].$$

Individuals with outlier values for a phenotype were removed from association analyses for that phenotype.

2.5 | Covariates

2.5.1 | Age, sex, and body mass index (BMI)—Biometric covariates such as age, sex, BMI, and height were measured directly at time of recruitment. BMI was categorized into underweight, normal, overweight, and obese, according to CDC guidelines for defining childhood obesity (Barlow, 2007; Cote et al., 2013; Whitlock et al., 2005). An overweight status was defined as a BMI at or above the 85th percentile for the general population of children of the same sex and in the same age group. An obese status was defined as a BMI at or above the 95th percentile. Underweight individuals (bottom 5th percentile, $n = 9$) were excluded from analysis.

2.5.2 | Asthma status—Case status was defined as physician-diagnosed asthma supported by reported asthma medication use and symptoms of coughing, wheezing, or shortness of breath in the 2 years preceding enrollment.

2.5.3 | Maternal educational attainment—Maternal educational attainment was measured at recruitment and included in analyses to control for socioeconomic status. It was coded as total years of education completed from the first grade: for example, a complete K-6 education was 6 years, a complete high school education was 12 years, and any additional years (college or trade school and beyond) were counted as 1 year each.

2.5.4 | Genetic ancestry—Previous literature on the genetics of non-Hispanic African Americans has observed global genetic ancestry proportions of 73.2% West African, 24.0% European, and 0.8% Amer-indigenous (Bryc et al., 2015), strongly suggesting that a reference panel of African and European was sufficient for accurate global genetic ancestry estimates in SAGE. Global genetic ancestry was estimated for each individual with the ADMIXTURE software (Alexander et al., 2009) in supervised learning mode assuming one West African and one European ancestral population, with HapMap Phase III YRI and CEU populations as references (The International HapMap 3 Consortium, 2010). Local ancestry estimation was performed with RFMix (Maples et al., 2013; Spear et al., 2019) using the same two-way ancestry reference from HapMap Phase III.

2.5.5 | Estimation of genetic relatedness and genotype principal components

—Genetic relatedness matrices (GRMs) were generated in R using GENESIS (Gogarten et al., 2019), which provides a computational pipeline for handling complex population structure using principal components analysis (PCA). We used PCAir (Conomos et al., 2015) to correct for distant population structure accounting for relatedness, and PC-Relate (Conomos et al., 2016) to adjust for genetic relatedness in recently admixed populations. The resulting principal components provide better correction for population stratification in admixed populations compared to standard PCA on genotypes (Patterson et al., 2006). Additionally, since GRMs produced by GENESIS are corrected for cryptic relatedness, the resulting association test statistics do not suffer inflation resulting from confounding relatedness in our sample.

2.6 | Genetic association analyses

Genotype association testing was performed with the MLMA-LOCO algorithm from GCTA (Yang et al., 2011, 2014) to correct for population structure using GRMs generated with GENESIS. Association testing of outcome phenotypes with allele dosages at 15,954,804 biallelic SNPs was performed with a “leave one chromosome out” model to avoid double-fitting tested variants. Other variables included in models were age, sex, BMI, maternal educational attainment, and three genotype principal components. Models of pre-BD also included asthma status.

The suggestive and significant association thresholds for each outcome phenotype were determined by the effective number of independent statistical tests (M_{eff}) calculated with CODA (Plummer et al., 2006). CODA computes M_{eff} using the autocorrelation of p values from GWAS. This produces population-specific Bonferroni thresholds that account for correlation between statistical tests without increasing the Type I error rate (Sobota et al., 2015). For our analyses, M_{eff} ranged from 488,819 to 507,975 (Table S2). The Bonferroni corrected genome-wide significance threshold was computed as $0.05/M_{\text{eff}}$, while the suggestive threshold was computed as $(1/M_{\text{eff}})$, yielding a single pair of thresholds for all six outcome phenotypes considered: $p < 1.99 \times 10^{-6}$ for suggestive association, and $p < 9.95 \times 10^{-8}$ for significant association (Table S2).

Admixture mapping analyses were performed using linear regression models in R and local ancestry calls from RFMix for 454,322 genotyped SNPs. Counts of 0, 1, or 2 alleles of African descent were computed for each person at each SNP. Phenotypes were then regressed onto ancestral allele counts for each SNP while including age, sex, height, BMI, maternal educational attainment (as a proxy for socioeconomic status), and global African genetic ancestry proportion as covariates. Analyses with pre-BD outcome measures also included asthma status as a covariate.

2.7 | Fine-mapping genetic associations

Functional fine-mapping with PAINTOR (Kichaev et al., 2014) was used to identify putative causal variants in novel loci deemed statistically significant by admixture mapping. PAINTOR applies a Bayesian probabilistic framework to integrate functional annotations, association summary statistics (Z -scores), and linkage disequilibrium information for each

locus to prioritize the most likely causal variants in a given region. Functional annotations were selected per locus as recommended by the authors of PAINTOR (Kichaev, 2017). A subset of lung- and blood-related functional annotations from the Roadmap Epigenomics Project (Roadmap Epigenomics Consortium et al., 2015) and the ENCODE Consortium (ENCODE Project Consortium, 2012) were assessed for their individual improvement to the posterior probability of causality; the top five minimally correlated annotations were selected for each locus.

2.8 | Annotation tools

The NHGRI/EBI GWAS Catalog (Buniello et al., 2019), Ensembl Genome Browser release 98 (Cunningham et al., 2019) and gnomAD browser v3.0 (Karczewski et al., 2020) were used to look up known associations at significant loci according to our analyses. Annotation lookups in the gnomAD browser v3.0 used hg38 coordinates translated from our hg19-aligned genotypes via liftOver (Hinrichs et al., 2006). Data management, statistical analysis, and figure generation made extensive use of GNU parallel (Tange, 2018) and several R packages, including data.table, doParallel, optparse, ggplot2, and the tidyverse bundle (Davis, 2020; Dowle & Srinivasan, 2020; Microsoft Corporation & Weston, 2019; Wickham, 2016; Wickham & Garrett, 2017).

3 | RESULTS

3.1 | Cohort characteristics

Characteristics of all SAGE participants included in analyses are shown in Table 1. Distributions of each lung function measure stratified by case-control status and BD administration (pre-BD vs. post-BD) are shown in Figure S1. FVC showed no significant difference between asthma cases and controls (Kruskal-Wallis $p = .073$), while stratification by case-control status yielded significantly different distributions for FEV₁ (Kruskal-Wallis $p = 4.8 \times 10^{-7}$) and FEV₁/FVC (Kruskal-Wallis $p = 1.5 \times 10^{-7}$). Among cases, statistically significant differences were observed between distributions of pre-BD and post-BD measures of FEV₁ (Kruskal-Wallis $p = 1.2 \times 10^{-38}$), FVC (Kruskal-Wallis $p = 5.4 \times 10^{-16}$), and FEV₁/FVC (Kruskal-Wallis $p = 4.0 \times 10^{-29}$), illustrating a measurable effect of BD medication on lung function.

The global African genetic ancestry proportion in our sample varied from 30.7% to 100%, with an average proportion of 80.2% (Figure S2), concordant with empirically observed averages (Baharian et al., 2016). Global ancestry contained the same information as the first genotype principal component ($R^2 = 0.984$, Figure S3).

3.2 | Genetic association testing finds novel and known loci

Figure 1 shows results from GWAS performed on pre-BD phenotypes (pre-FEV₁, pre-FVC, and pre-FEV₁/FVC) and post-BD phenotypes (post-FEV₁, post-FVC, and post-FEV₁/FVC) using linear mixed modeling. The association results showed no evidence of genomic inflation, with genetic control λ ranging from 0.98 to 0.99 (Table S2 and Figure S4). Table 2 lists the 18 genome-wide significant associations found, each associated with exactly one of the six lung function measures. An additional 252 variants were suggestively associated with

at least one phenotype (Tables S3–S8). Of the 18 variants, 4 variants on chromosome 13 in a region spanned by the gene *ATP8A2* were associated with pre-FEV₁/FVC (Figure S6). Two variants on chromosome 16 that were associated with pre-FVC flanked the promoter region of *IRX3* (Figure S7). A third variant associated with pre-FVC was located on chromosome 20 near *THBD* (Figure S8), a gene linked to venous thromboembolism in African American and Afro-Caribbean individuals (Hernandez et al., 2016). Two variants associated with Post-FVC were in a gene-rich region on chromosome 19 (Figure S9), with the peak near *TMIGD2* and *SHD*, while eight other variants pointed to a second gene-rich region on chromosome 11 near *CXCR5* and *HYOU1* (Figure S10). Post-FEV₁/FVC was associated with a region on chromosome 15 near the genes *AKAP13* and *ADAMTS7P4* (Figure S11).

Among the suggestive associations, a variant on chromosome 12 associated with post-FEV₁ was near *BTBD11* (Figure S12), a gene previously associated with post-FEV₁, post-FEV₁/FVC, and Δ FEV₁, the change in lung function due to BD administration (Hardin et al., 2016; Lutz et al., 2015), as well as BMI (Kichaev et al., 2019). A suggestive association with pre-FEV₁ on chromosome 12 fell near *SCARB1* (Figure S13), which was previously associated with FEV₁ and FVC (Wyss et al., 2018) and HDL cholesterol levels (Wojcik et al., 2019). Another suggestive association with pre-FEV₁ on chromosome 20 was near the gene *PTPRT* (Figure S14), which was previously associated with thromboembolism susceptibility in 5334 African American individuals (Heit et al., 2017).

3.3 | Admixture mapping identified five novel loci not found by GWAS

Table 3 shows five regions where admixture proportions were statistically significantly associated with one of the six phenotypes. The three pre-BD phenotypes (pre-FEV₁, pre-FVC, pre-FEV₁/FVC) were each associated with one region, while post-FVC was associated with two distinct regions. Post-FEV₁ and post-FEV₁/FVC had no significant associations. None of the regions overlapped with those significant in our GWAS, and none showed large deviations from mean genome-wide African genetic ancestry. A small region on chromosome 21 that was significantly associated with pre-FEV₁ flanked the genes *ADAMTS1* and *ADAMTS5* (Figure S15). The region on chromosome 4 associated with pre-FVC pointed to two candidate genes, *RCHY1* and *THAP6*, that had no prior lung disease associations (Figure S17). A region on chromosome 19 associated with pre-FEV₁/FVC spanned the genes *ZNF557* and *INSR* (Figure S18). Post-FVC was associated with two regions, one on chromosome 8 spanning the genes *ESRP1*, *INTS8*, *TP53INP1*, and *NDUFA6* (Figure S19), and another on chromosome 14 encompassing *EGLN3* and *SNX6* (Figure S20).

3.4 | Functional fine-mapping found three novel putatively causal loci for lung function phenotypes

Table 4 lists the most probable causal SNP for each of the five admixture mapping loci according to PAINTOR. SNP rs13615 showed the highest probability of causality (0.630) with pre-FEV₁ on locus 1 (Figure 2). This variant falls within the 3'-untranslated region (3'-UTR) of *ADAMTS1*, suggesting that *ADAMTS1* drives the admixture mapping association and not its physical neighbor *ADAMTS5*. The MAF of rs13615 in African and African diaspora populations was lower than every other global population (2.6% AFR vs. 7.0%–

54.5% other populations, gnomAD v3; see Figure S21). The SNP rs10857225 emerged as the most likely causal variant (probability 0.361) for the association of pre-FVC with locus 2 on chromosome 4 (Figure 3). This variant is located within an intron of the gene *THAP6*, suggesting that *THAP6* is more likely the causal gene behind the association with pre-FVC. In contrast to locus 1, the MAF of rs10857225 is highest in global African populations and markedly lower in other global populations (59.1% AFR vs. 28.1%–38.4% other populations, gnomAD v3). Locus 3 on chromosome 19 associated with pre-FEV₁/FVC, and locus 4 on chromosome 8 associated with post-FVC, showed little information gain from functional fine-mapping. The driving variant for locus 3, SNP rs72986681, was located in the 3'-UTR of *ZNG557*, but showed a low probability of causality (0.168, Figure S22). The most probable marker for locus 4, the SNP rs2470740, which is located in intron 2 of *RAD54B*, showed an even lower probability of causality (0.109, Figure S23). Functional fine-mapping of locus 5, a region on chromosome 14 associated with post-FVC, yielded the SNP rs1351618 with a moderate probability of causality (.390, Figure 4). rs1351618 is located in an intron of *EGLN3*. As with locus 2, rs1351618 had a much higher MAF in populations of African ancestry versus other global populations (12.4% AFR vs. <2.2% other populations, gnomAD v3).

4 | DISCUSSION

We analyzed the genetic basis of six lung function phenotypes in 1103 non-Hispanic African American children with and without asthma. The phenotypes consisted of three standard spirometric measures—FEV₁, FVC, and FEV₁/FVC—measured before and after administration of BD medication. Our GWAS identified 18 genome-wide significant loci, while our integrative genetic analysis approach that layered GWAS, admixture mapping, and functional fine-mapping identified another five putatively causal loci that could drive differences between pre- and post-BD lung function.

The four variants on chromosome 13 associated with pre-FEV₁/FVC pointed to *ATP8A2* as a candidate gene. *ATP8A2* encodes an ATPase involved in phospholipid transport and is highly expressed in brain tissue, testes, and the adrenal glands, and to a lesser degree in the lung (Fagerberg et al., 2014). Mutations in *ATP8A2* have been linked to several neurological disorders (Martín-Hernández et al., 2016). Two variants on chromosome 16 that were associated with pre-FVC point to *IRX3* as a candidate gene. *IRX3* encodes a homeobox protein crucial for neural development, and its promoters previously showed a long-range interaction with the *FTO* gene. Expression levels of the *FTO* gene are known to influence BMI and are of great interest in type II diabetes and obesity research (Smemo et al., 2014). Post-FVC showed associations on two chromosomes. Notable genes near the association peak on chromosome 19 included *MAP2K2* and *ZBTB7A*, genes associated with variation in the two genes closest to the association peak, *TMIGD2* and *SHD*, have not been previously associated with any traits. *TMIGD2* is involved in T-cell costimulation and the immune response through an interaction with BMI, visceral adiposity, and eosinophil counts (Kichaev et al., 2019; Pulit et al., 2019; Rüeger et al., 2018); and *CHAF1A*, *HDGFL2*, *PLIN4*, *ANKRD24*, *MPND*, and *SH3GL1*, previously associated with corpuscular volume and hemoglobin concentration (Astle et al., 2016; Kichaev et al., 2019; van Rooij et al., 2017). Interestingly, *HHLA2*, suggesting that it could possibly play an immune or allergic

response role in lung function (Y. Zhu et al., 2013). Among the genes within or near the chromosome 11 peak, two emerge as potentially key loci. The first is *CXCR5*, which has been linked to increased risk of childhood onset asthma (Ferreira et al., 2019; Johansson et al., 2019; Pividori et al., 2019) and respiratory disease (Kichaev et al., 2019), as well as related allergic and immunological conditions such as eczema, leukocyte count, rheumatoid arthritis, and Sjögren's syndrome (Ferreira et al., 2017; Kichaev et al., 2019; Laufer et al., 2019; Lessard et al., 2013). The second is *HYOU1*, which has been associated with BMI and post-FEV₁/FVC (Lutz et al., 2015; Pulit et al., 2019). Post-FEV₁/FVC was associated with two genes, *AKAP13* and *ADAMTS7P4*. *AKAP13* has been previously associated with numerous conditions, including interstitial lung disease and psoriasis in European individuals (Fingerlin et al., 2013; Tsoi et al., 2015) as well as weight, BMI, and cardiovascular traits such as blood pressure and hemoglobin count in multiple populations (Giri et al., 2019; Kichaev et al., 2019). *ADAMTS7P4* was previously associated with red blood cell volume (Kichaev et al., 2019). The statistically suggestive association of post-FEV₁ with *BTBD11* pointed to previous associations with various lung function measures, including post-FEV₁, post-FEV₁/FVC, and Δ FEV₁ (Hardin et al., 2016; Lutz et al., 2015). These previously detected associations were based on much larger sample sizes than what was available to us: the associations with post-FEV₁ and post-FEV₁/FVC found by Lutz et al. were discovered in a population of 10,094 European and 3260 African American smokers with chronic obstructive pulmonary disorder (COPD), while the association with Δ FEV₁ found by Hardin et al. was based on 5766 Europeans and 811 African Americans with COPD, suggesting that our inability to reach genome-wide significance in our sample was due to insufficient statistical power.

Among significant and suggestive GWAS loci, the association of post-FVC with variants in or near *CXCR5* and *HYOU1* is the only one that replicates known lung function loci: *CXCR5* was previously associated with asthma (Ferreira et al., 2019; Johansson et al., 2019; Pividori et al., 2019), and *HYOU1* was previously associated with post-FEV₁/FVC (Lutz et al., 2015). The association with post-FEV₁/FVC comes from an adult COPD cohort ascertained by smoking status; in contrast, SAGE is a pediatric asthma cohort. The mechanism by which *HYOU1* affects lung function in both youth and adults is unclear. Nevertheless, the overlap of post-BD pulmonary function measures at this locus suggests that the region encompassing *CXCR5* and *HYOU1* plays a role in lung disease among people with obstructive lung function.

Admixture mapping identified five genomic regions where variation in genetic ancestry was significantly associated with phenotypic variation. Locus 1 on chromosome 21 spanned the genes *ADAMTS1* and *ADAMTS5*, which encode extracellular proteases within the same protein family but with different consequences for disease. Although both genes have been linked to blood protein levels (Suhre et al., 2017), *ADAMTS1* has been associated with pre-FVC (Kichaev et al., 2019) and is expressed in arterial, adipose, and lung tissue, while *ADAMTS5* is not appreciably expressed in the lung (Figure S16). Further fine-mapping with PAINTOR places the most likely causal SNP (rs13615) within the 3'-UTR of *ADAMTS1*. Although the region is sparsely genotyped, and follow-up with whole genome sequencing data in this region is recommended, these results suggest that *ADAMTS1* may be functionally related to lung function. Interestingly, the association of *ADAMTS1* with pre-

FVC (Kichaev et al., 2019) was discovered in a European sample of substantially larger size than our cohort, highlighting the ability of admixture mapping to detect associations in scenarios with low statistical power. Locus 3, spanning a region on chromosome 19 that was associated with pre-FEV₁/FVC, contains the genes *ZNF557* and *INSR*. *ZNF557* has not been previously associated with any traits, while *INSR* is the well-known insulin receptor that has been previously associated with childhood onset asthma in our own cohort (White et al., 2016), as well as blood pressure levels, triglyceride levels, HDL cholesterol levels, and hypothyroidism across multiple populations (Bentley et al., 2019; Ehret et al., 2016; Kichaev et al., 2019; Klarin et al., 2018). Post-FVC showed two distinct admixture mapping signals. The first region on chromosome 8, which we call Locus 4, includes the genes *ESRP1*, *INTS8*, *TP53INP1* and *NDUFAF6* was previously associated with type II diabetes and eosinophil counts (Kichaev et al., 2019; Mahajan et al., 2018). The second region, Locus 5, spans *EGLN3* and *SNX6*, both of which show previous associations with blood phenotypes such as blood pressure and hematocrit levels (Astle et al., 2016; Evangelou et al., 2018).

Overall, evaluation of our GWAS and admixture mapping lung function results suggests that genetics of this trait underlie some pleiotropy observed across pulmonary, hematological, cardiovascular, and obesity-related traits. Such pleiotropy has been observed in UK BioBank participants: as lung function decreases, BMI and type II diabetes incidence increases, as well as levels of eosinophils and neutrophils, both of which are common biomarkers for allergic disease (Figure S25; McInnes et al., 2019). The link between obesity and lung function is particularly interesting since obesity is a known asthma comorbidity, and lung function may play a role in obese asthma (Baffi et al., 2015; Gruchała-Niedoszytko et al., 2015). Our findings suggest that genetically based differences in lung function may provide a link between obesity and asthma.

It is curious that the regions identified by admixture mapping and subjected to functional fine-mapping did not overlap with the statistically significant GWAS loci. We attribute this in part to the different types of information used by each approach: GWAS analyzes how allelic variation affects a trait, while admixture mapping analyzes the phenotypic consequences of variation in genetic ancestry. Our integrative approach deprioritized results solely from GWAS, an approach driven by the fact that a supermajority of individual GWAS results fall in noncoding regions of the genome and are consequently notoriously difficult to interpret. By integrating GWAS summary statistics with loci identified via admixture mapping, we found that three of the admixture mapping-based loci—*ADAMTS1*, *THAP6*, and *EGLN3*—had evidence of causal effects. Each of the sentinel SNPs tagging these genes showed a notable difference in ancestral allele frequency: populations of African descent had either the highest or the lowest MAF among all global populations, likely the result of admixture mapping prioritizing loci that varied by genetic ancestry. None of these loci have been previously associated with lung traits, highlighting the strength of our integrative analysis. The association with *EGLN3* is particularly curious since it has been previously associated with a variety of traits, including heart rate response to β -blocker therapy (Shahin et al., 2018). Short-acting β -2 agonists such as albuterol selectively target β -2 receptors in the lungs, while the first-generation β -blockers taken for cardiac conditions bind to both β -1 and β -2 receptors, affecting the heart as well the lungs. Bronchospasm and FEV₁ reduction are clinically significant side effects of first-generation β -1 selective and nonselective β -

blockers for cardiac conditions. Consequently, these nonselective β -blockers must be initiated with caution and close monitoring in patients with asthma (Christiansen & Zuraw, 2019). β -blockers lower blood pressure by reducing heart rate and cardiac contractility and are less effective in people with high levels of African genetic ancestry (Brewster & Seedat, 2013; Whelton et al., 2018). It has been previously observed that African Americans with asthma demonstrate lower BD drug response than European Americans (Blake et al., 2008), suggesting a possible pharmacological interaction between β -2 receptors and African ancestry. Furthermore, *EGLN3* is strongly expressed in cardiac tissue, suggesting that *EGLN3* could possibly influence post-FVC through cardiac phenotypes (Figure S24). Further functional studies are required to elucidate the role of *EGLN3* on lung function and BD drug response.

This study has some important limitations. First, while our data set includes rich phenotyping of pulmonary traits with socioeconomic and biometric measures, it still constitutes a somewhat small sample by modern measures. The tradeoff between rich phenotyping and increased sample size is not trivial, particularly for studies of populations traditionally underrepresented in genetic research, a challenge that still plagues large studies like NHLBI TOPMed and NIH Million Veterans Program. Our approach of layering multiple types of genomic information serves as a partial workaround. However, by layering GWAS with admixture mapping and functional fine-mapping, we restricted our focus to regions where variants showed differential ancestry, which excluded our strongest GWAS hits. While PAINTOR can provide evidence of putatively causal markers that do not necessarily meet strict genome-wide thresholds of significance, it was not designed with admixed populations in mind. We applied suitably rigorous statistical stringency to both our GWAS and our admixture mapping results by thresholding to the effective number of independent tests as estimated by the CODA software, an approach designed to produce population-specific significance thresholds (Sobota et al., 2015), so we are confident that these regions are indeed correlated to the phenotype. Nevertheless, further research is needed to understand how the hierarchical Bayesian model in the PAINTOR inference engine behaves in regions of heterogeneous ancestry. Finally, if the truly causal variant is not genotyped, then PAINTOR and similar fine-mapping frameworks suffer a performance hit that is difficult to rectify without ultra-fine genotyping in the locus of interest (e.g., from whole genome sequencing). We suspect that our relatively low probabilities of causality can be attributed to a confluence of differential ancestry, small sample size, and insufficiently resolved fine-mapping, which could be improved in future studies by using whole-genome sequencing data instead of imputed genotypes.

Our integrative analysis approach leverages available functional annotations and genetic ancestry estimates in the absence of molecular data to yield some promise for discovery of novel loci. Our study is limited to three tiers—genotypes, genetic ancestry, and functional annotations—and makes use of gene expression results from GTEx v8. However, it does not directly incorporate any transcriptomic, metabolomic, proteomic, or methylomic information. As large multiomic data sets from NHLBI TOPMed, UK Biobank, and the NIH Million Veterans Program become available, the need for integrative genomic approaches to studying complex diseases will increase. Future multiomic models of complex diseases, including obstructive lung function disorders, may deliver on the promise of precision

medicine and provide actionable clinical translation of biomedical and pharmacogenomic insights into novel therapies.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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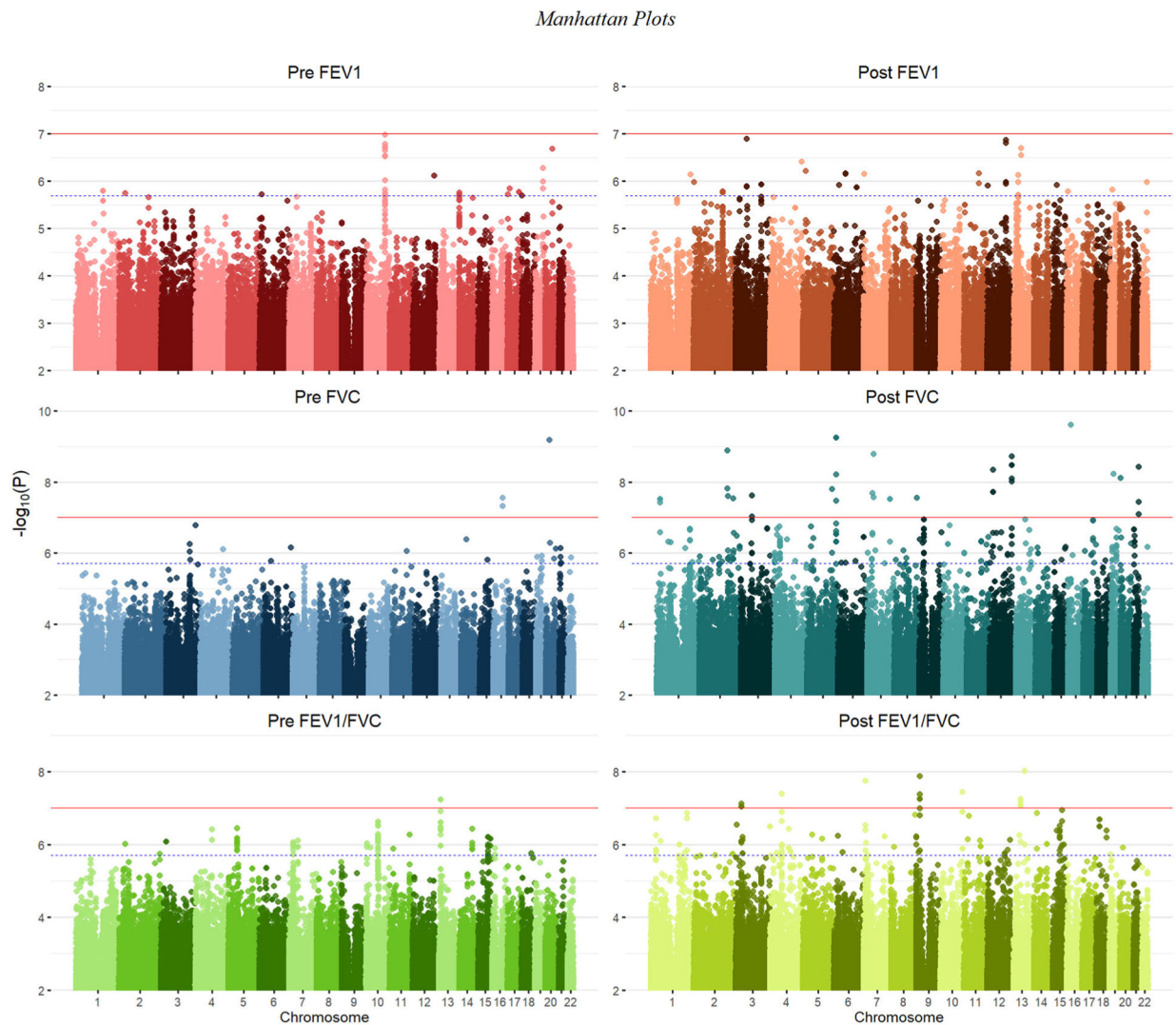
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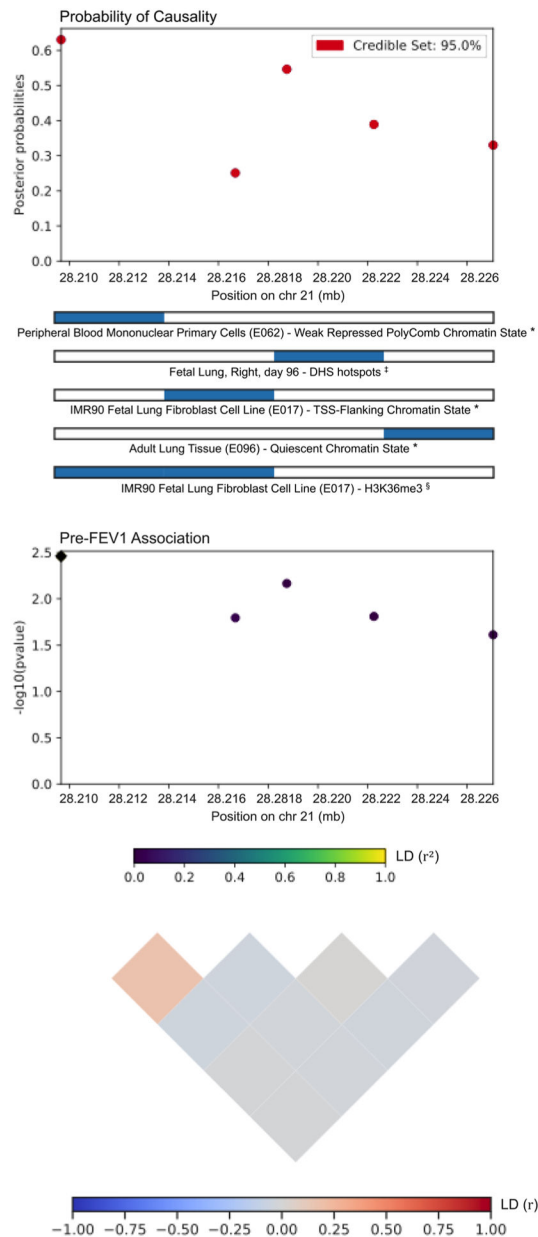
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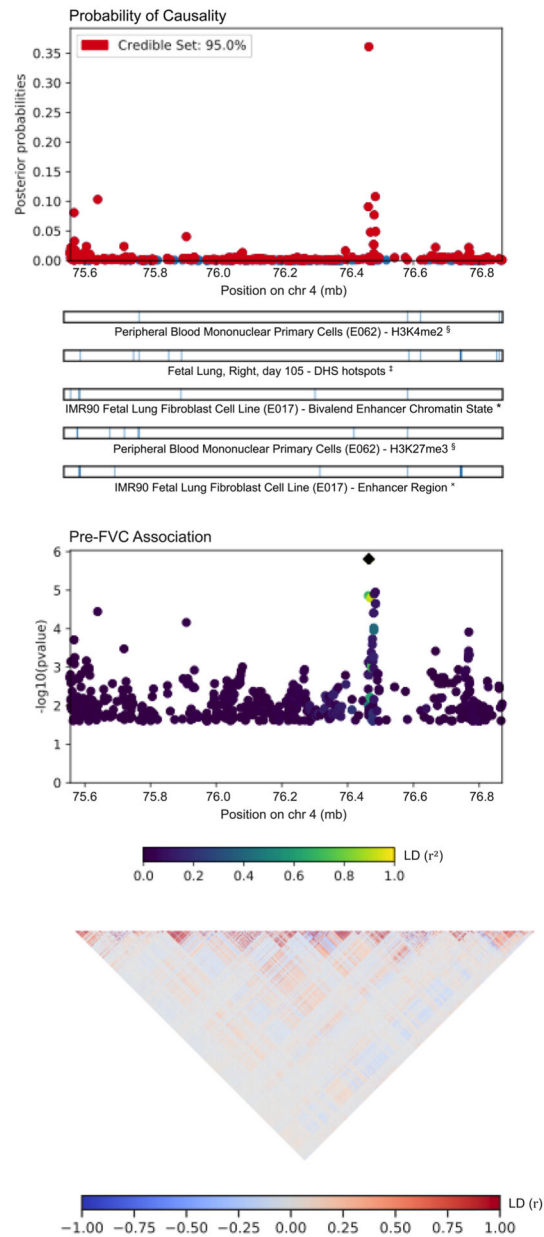
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**FIGURE 1.**

Manhattan plots summarizing GWAS p values for all six lung function phenotypes. The solid red line denotes genome-wide significance ($p < 9.95 \times 10^{-8}$), while the dashed blue line marks the suggestive threshold ($p < 1.99 \times 10^{-6}$), per CODA calculations. Variants with a p value greater than 0.05 were deemed uninformative and therefore not plotted

**FIGURE 2.**

A CANVIS plot of results from PAINTOR functional fine-mapping for locus 1, an association with pre-FEV₁ on chromosome 21. The SNP rs13615, which sits in the 3'-UTR of the gene *ADAMTS1*, attains a posterior probability of causality of 0.630. The panels show, from top to bottom, the posterior probability of causality; the five most informative functional annotations; GWAS *p* values; and local linkage disequilibrium expressed as a signed Pearson correlation. 3'-UTR, 3'-untranslated region; FEV₁, forced expiratory volume in 1 s; SNP, single nucleotide polymorphism

**FIGURE 3.**

PAINTOR results for locus 2, an association on chromosome 4 with pre-FVC. The sentinel SNP, rs10857225, corresponds with a GWAS peak that does not pass Bonferroni correction for statistical significance. The highlighted peak tags the intron of the gene *THAP6*. FVC, forced vital capacity; SNP, single nucleotide polymorphism

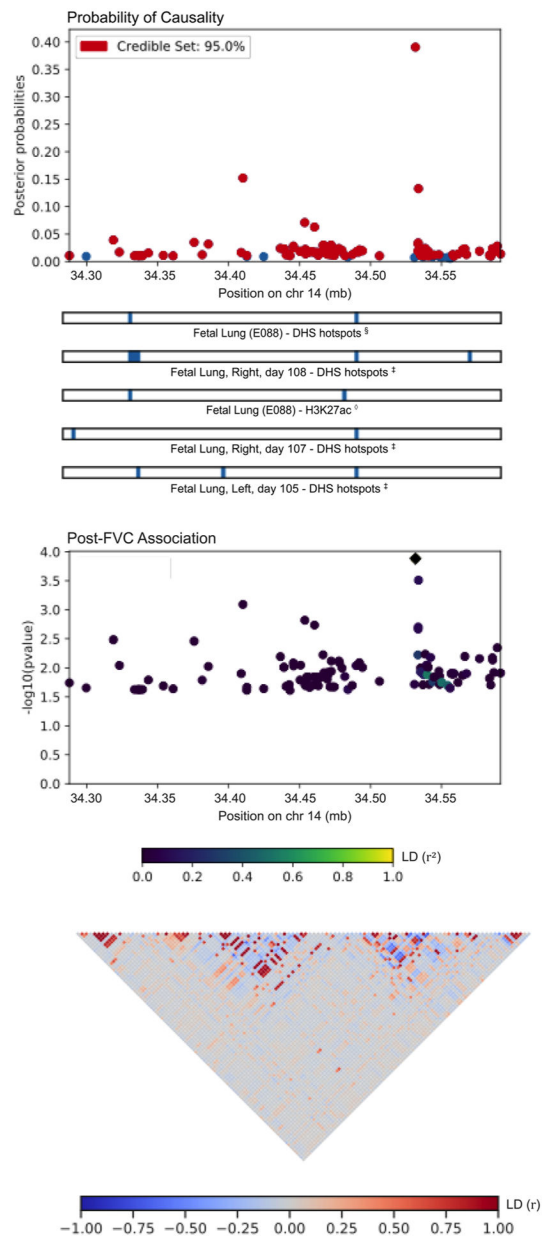


FIGURE 4. PAINTOR fine-mapping results for locus 5, corresponding to a region on chromosome 14 associated with post-FVC. The most likely causal SNP, rs1351618, tags an intron of the gene *EGLN3*. FVC, forced vital capacity; SNP, single nucleotide polymorphism

TABLE 1

Summary statistics of phenotypes and covariates from the SAGE cohort

Characteristics	Cases	Controls	Total
Subjects (<i>n</i>)	831	272	1,103
Age (year)	14.1 (3.66)	16.3 (3.77)	14.7 (3.8)
Female (<i>n</i>)	406	166	572
Height (cm)	158 (14.34)	162.4 (13.26)	159.1 (14.2)
African ancestry (%)	80.4 (0.1)	79.6 (0.1)	80.2 (0.1)
Maternal education (yr)	12.4 (1.47)	12.2 (1.5)	12.3 (1.48)
Obesity status			
Obese (<i>n</i>)	276	74	350
Nonobese (<i>n</i>)	555	198	753
Pre-FEV ₁	103 (13.79)	98.1 (13.02)	99.3 (13.77)
Pre-FVC	103.4 (12.84)	105.1 (13.09)	103.8 (12.92)
Pre-FEV ₁ /FVC	95.1 (9.35)	98.4 (8.2)	95.9 (9.19)
Post-FEV ₁	107 (13.44)	n/a	n/a
Post-FVC	109 (14.42)	n/a	n/a
Post-FEV ₁ /FVC	99 (7.83)	n/a	n/a

Note: Displayed numbers are either counts (*n*) or averages followed by standard errors in parentheses. Units are listed where appropriate. An “n/a” appears where measurements were taken on cases only

Abbreviations: FEV₁, forced expiratory volume in 1 s; FVC, forced vital capacity.

TABLE 2

Significant association results from GWAS

Phenotype	Chr	Position (bp)	A1	A2	MAF	β	SE	p value	Genes
Post-FEV1/FVC	15	85798401	A	G	0.018	-6.7846	1.27	9.05×10^{-8}	<i>ADAMTS74P, AKAP13</i>
Post-FVC	11	118932913	C	T	0.010	18.9094	3.29	9.47×10^{-9}	<i>CXCR5, HY0U1</i>
Post-FVC	11	118902275	C	T	0.010	17.9775	3.29	4.83×10^{-8}	<i>CXCR5, HY0U1</i>
Post-FVC	11	118905095	A	G	0.010	17.9775	3.29	4.83×10^{-8}	<i>CXCR5, HY0U1</i>
Post-FVC	11	118905316	T	C	0.010	17.9775	3.29	4.83×10^{-8}	<i>CXCR5, HY0U1</i>
Post-FVC	11	118906065	A	G	0.010	17.9775	3.29	4.83×10^{-8}	<i>CXCR5, HY0U1</i>
Post-FVC	11	118906240	T	C	0.010	17.9775	3.29	4.83×10^{-8}	<i>CXCR5, HY0U1</i>
Post-FVC	11	118906745	C	G	0.010	17.9775	3.29	4.83×10^{-8}	<i>CXCR5, HY0U1</i>
Post-FVC	11	118907923	G	T	0.010	18.2978	3.41	7.85×10^{-8}	<i>CXCR5, HY0U1</i>
Post-FVC	19	4289259	T	C	0.106	5.1672	0.967	9.21×10^{-8}	<i>TMIGD2, SHD</i>
Post-FVC	19	4291817	T	C	0.111	5.2674	0.963	4.52×10^{-8}	<i>TMIGD2, SHD</i>
Pre-FEV1/FVC	13	26235394	G	A	0.010	-10.5334	1.94	5.80×10^{-8}	<i>ATP8A2</i>
Pre-FEV1/FVC	13	26247080	G	A	0.010	-10.5334	1.94	5.80×10^{-8}	<i>ATP8A2</i>
Pre-FEV1/FVC	13	26262378	T	G	0.011	-10.5325	1.94	5.81×10^{-8}	<i>ATP8A2</i>
Pre-FEV1/FVC	13	26268604	A	C	0.011	-10.5325	1.94	5.81×10^{-8}	<i>ATP8A2</i>
Pre-FVC	20	22900228	G	A	0.010	18.0953	2.93	6.77×10^{-10}	<i>THBD</i>
Pre-FVC	16	54327903	G	A	0.023	9.8543	1.78	2.83×10^{-8}	<i>IRX3, FTO</i>
Pre-FVC	16	54327610	A	G	0.034	8.3689	1.53	4.85×10^{-8}	<i>IRX3, FTO</i>

Note: The p values for all SNPs listed here met the significance threshold of 9.95×10^{-8} . SNPs were specified by chromosome (Chr) and physical position in base pairs (bp).

Abbreviations: A1, major allele; A2, minor allele; B, effect size; FEV1, forced expiratory volume in 1 s; FVC, forced vital capacity; Genes, any genes within proximity of the associated variants; MAF, minor allele frequency; SE, standard error of the estimate of β .

TABLE 3
Admixture mapping in SAGE identified five regions with statistically significant association to at least one phenotype

Locus	Phenotype	Chr	Start (bp)	End (bp)	Length (kb)	Threshold	SNP _{Admix} (n)	SNP _{GWAS} (n)	AFR (%)	Genes
1	Pre-FEV1	21	28,209,667	28,240,392	30.72	1.03×10^{-4}	4	215	78.7	<i>ADAMTS1</i>
2	Pre-FVC	4	75,555,658	76,873,740	1318.08	9.64×10^{-5}	102	7905	79.8	<i>RCHY1, THAP6</i>
3	Pre-FEV1/FVC	19	7,068,207	7,127,294	59.09	1.04×10^{-4}	18	376	80.4	<i>INSR, ZNF557</i>
4	Post-FVC	8	95,387,941	95,820,594	432.65	9.93×10^{-5}	45	2822	80.8	<i>ESRPI, INTS8, TP53NP1, NDUFAF6</i>
5	Post-FVC	14	34,283,561	34,595,061	311.50	9.93×10^{-5}	95	1982	80.5	<i>EGLN3, SNX6</i>

Note: The regions are arbitrarily numbered from 1 to 5 and defined by phenotype, chromosome (Chr), physical starting point (in base pairs), and end point (in base pairs). Physical positions are given in hg19 coordinates. The total length of the region is given in kilobasepairs (kb). The threshold for statistical significance is given for each region. The column “Genes” lists genes physically within and near the associated regions.

Abbreviations: AFR, the percentage (%) of local genetic ancestry of African origin; FEV₁, forced expiratory volume in 1 s; FVC, forced vital capacity; SNP, single nucleotide polymorphism; SNP_{Admix}, counts the number (n) of genotyped SNPs from admixture mapping that met the significance threshold; SNP_{GWAS}, the total number of GWAS SNPs (genotyped and/or imputed) in the admixture mapping region.

TABLE 4
Results from PAINTOR highlighting the most probably causal SNPs for each locus as defined by admixture mapping

Locus	Phenotype	Chr	Position (bp)	SNP	Ref allele	Alt allele	MAF (%)	P-value	Annotation	Pr (causal)
1	Pre-FEV ₁	21	28,209,667	rs13615	A	G	2.55	6.95×10^{-3}	<i>ADAMTS1</i> , 3'-UTR	0.630
2	Pre-FVC	4	76,464,584	rs10857225	A	C	59.07	3.11×10^{-6}	<i>THAP6</i> , Intron Variant	0.361
3	Pre-FEV ₁ /FVC	19	7,087,789	rs72986681	G	A	1.54	7.25×10^{-4}	<i>ZNF557</i> , 3'-UTR	0.168
4	Post-FVC	8	95,399,551	rs2470740	A	T	23.82	2.04×10^{-3}	<i>RAD54B</i> , Intron Variant	0.109
5	Post-FVC	14	34,531,633	rs1351618	C	T	12.40	1.99×10^{-4}	<i>EGLN3</i> , Intron Variant	0.390

Note: As in Table 3, the loci are arbitrarily numbered from 1 to 5 and defined by phenotype and chromosome. The physical position (in basepairs) of the most likely causal SNP is given in hg19 coordinates. MAFs are taken from global populations from the gnomAD server v3. The displayed *p* values are from our discovery GWAS. Pr(causal) is computed from PAINTOR.

Abbreviations: 3'-UTR, 3'-untranslated region; Chr, chromosome; FEV₁, forced expiratory volume in 1 s; FVC, forced vital capacity; MAF, minor allele frequency; Pr (causal), probability of causality; SNP, single nucleotide polymorphism.